Decision Log — Knotted Coil Designer

Dated entries: what was decided, why, who decided it, and what the alternatives were. Newest first.

2026-10-20 — Field maps (B1 / E) requested; not built yet

- **Request (Reza):** a UI toggle to visualise E-fields.
- **Claude's assessment:** a quasi-static wire model can compute the **magnetic field** properly, using Biot–Savart, and that can be checked against the analytic on-axis field of a loop. It can only give the **inductive part of E** (−∂A/∂t). The charge-driven part from capacitors and gaps, and the effect of the head, are missing, so an E map would be incomplete in a misleading way.
- **Status:** outside frozen Design v1.0 scope (§1, §8). **Awaiting Reza's decision.**

2026-10-20 — In-plane rotation as a solver option

- **Decided by:** Reza, who asked for rotation as a decoupling lever.
- **What changed:** rotation was already a free variable per element. It is now exposed as `rotation` = free / locked (one shared angle) / fixed.
- **Exploratory observation, one seed, N = 16, trefoil β = 0.4, nearest-ring target:** worst nearest-ring |k| was 0.010 with free rotation, 0.040 locked and 0.045 fixed.

2026-10-20 — Neighbour definition: frozen rule kept as default, 'ring1' added (PENDING Reza)

- **Finding:** a trefoil at β = 0.4 nulls only at a centre spacing of about 1.14 Rf, which is heavy overlap. At that spacing, second-ring elements (about 1.97 Rf in a hexagonal layout) still overlap.
- **Consequence:** under the frozen Design §5.1 definition (footprints overlap), the solver must null pairs that cannot be nulled together, and the problem is badly posed.
- **Option added:** `target = 'ring1'` decouples only the nearest ring (closer than 1.35 × the pair-null spacing).
- **Results (16 elements, exploratory):**
  - nearest ring: worst |k| ≈ 0.008–0.010;
  - second ring: |k| up to about 0.08.
- **Default is unchanged** (frozen definition). Every result reports **both** verdicts.
- **Decision needed from Reza:** which definition should be the default, i.e. which one "decoupled" should mean.

2026-10-20 — Validation runs 1 and 2: V6 FAIL (recorded), causes and fixes

- **Run 1** (git 9f89d91): V6 failed for three separate reasons.
  1. **Physics:** the bloom floor is above β = 0.02 for the placeholder wire (1 mm + 2 × 0.25 mm insulation) at R = 30 mm. Near the knot, the strands sit on a thin tube of in-plane radius βR. The p strands can be at most $2\beta R \cdot \sin(\pi/p)$ apart, so with c = d_w + 2t_i the minimum buildable bloom is $\beta \geq c / (2R \sin(\pi/p))$. That gives 0.025 for p = 2, 0.029 for p = 3 and 0.043 for p = 5. This is Claude's derivation; the numerical lift solver agreed (feasible at 0.03, 0.03 and 0.045).

2. **Discretisation:** 300 points missed one crossing of T(2,5) at β = 0.9 (600 or more resolve it). That shape also has a 0.08 mm bend radius, so it cannot be built.
3. **Harness bug:** the reference over/under pattern was taken at β = 0.02, which was itself the degenerate case.

- **Fixes:**
  - adaptive point count (`knot.adaptiveN`);
  - reference pattern taken at β = 0.4;
  - bloom floor shown in the UI.
  - The **criterion is unchanged.**
- **Run 2:** V6 still FAILS, now *only* at β = 0.02, for every knot. That is the physical floor.
- **Post-hoc check PH2 (informational, not a verdict):** topology is preserved from the floor upward for all catalogue knots.
- **Decision needed from Reza:** whether to change the β_min default to the wire-dependent physical floor. That is a Design change.

2026-10-20 — V3 reference is UNVERIFIED

- **Status:** Roemer et al. 1990 is paywalled; a subagent tried the Wiley PDF, Semantic Scholar, OpenAlex, Europe PMC and the Internet Archive.
- **Closest primary text read:** Roemer & Edelstein, US 4,825,162 (1989), “D approximately equal to 0.75d”. This is PARAPHRASE-ONLY, read from Google Patents OCR text.
- **V3 verdict:** NOT EVALUABLE. Our computed value is 0.7604 × diameter (exploratory comparison: +1.4% from the patent’s approximate value).
- **To resolve:** Reza retrieves the paper through McGill library access.

2026-10-20 — Kernel softening between elements

- **Choice:** wires of *different* elements use a $1/\sqrt{(r^2 + \varepsilon^2)}$ kernel with $\varepsilon = d_w + 2t_i$, so crossings between elements are treated as separated by the real centreline gap. This avoids having to model which wire goes over at each crossing.
- **Alternative:** explicit over/under layers per crossing.
- **Status:** sensitivity to ε not yet tested.

2026-10-20 — Design.md v1.0 frozen

- **Decided by:** Reza approved all §11 defaults.
- **Commit:** git 39a4e83; this commit pre-registers V1–V6.